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CSE13S

Assignment 6: Huffman Coding
Design Document

Purpose

The purpose of this assignment is to create a Huffman coding module with an encoder and a decoder using node, priority queue, code, and stack abstract data types. Huffman coding creates static encodings of information and allows for data compression. The optimal static encoding assigns the least number of bits to the most common symbol and the greatest number of bits to the least common symbol according to the entropy equation

$$H(X) = \sum_{i=1}^{n} \Pr[x_i] I(x_i) = -\sum_{i=1}^{n} \Pr[x_i] \log_2 \Pr[x_i].$$

for the set X with I(x) as the measure of the amount of information in the set.

Layout/Structure

- Encode.c-contains the implementation for the Huffman encoder
- Decode.c-contains the implementation for the Huffman decoder
- Entropy.c-provided by the resources repository and contains the code for measuring the entropy of a file
- defines.h - contains macro definitions for the assignment
- Header.h - contains the struct definition for a file header
- Node.h-provided by resource repository, contains the node ADT interface
- Node.c-contains the implementation for the node ADT
- pq.h -provided by resources repository, contains the priority queue ADT interface
- pq.c -contains the implementation of the priority queue ADT, priority queue struct is defined in this file
- Code.h- contains the code ADT interface
- Code.c-this file will contain your implementation of the code ADT
- Io.h- provided by resources repository. This file contains the I/O module interface.
- Io.c - contains the implementation for the I/O module
- stack.h - contains the stack ADT interface
- Stack.c -contains the implementation of the stack ADT, the stack struct is defined in this file
- Huffman.h-provided by the resources repository, contains the Huffman coding module interface
- Huffman.c-contains the implementation of the Huffman coding module interface
- Makefile-allows user to compile the programs

Node ADT

- The Huffman trees are made up of nodes.

- Each node has a pointer to its left child, a pointer to its right child, a symbol and a frequency of that symbol. The node's frequency is used by the decoder.
- A symbol is a uint8_t and the input is read as raw bytes. The definition of a node is made transparent.

Priority Queue ADT

- High priority elements are dequeued before low priority elements.
- Enqueue adds elements to a position based on priority.
- The priority queue will be implemented using an insertion sort to enqueue nodes.
- A lower frequency node has a higher priority.

Code ADT

- Represents a stack of bits
- The struct definition of a code is transparent so we can pass a struct by value
- The MAX_CODE_SIZE macro is defined in defines.h

I/O

- Uses low level syscalls like read(), write(), open(), and close()

Stacks

- The stack will be used to store nodes.

Functionality

Encoder

- The Huffman encoder reads from an input file, gets the Huffman encoding of the contents, and compresses the file using the encoding. It supports the command line options [-h] [-i infile] [-o outfile] [-v].
    - -h option prints out a help message the explains the purpose of the program and the command line options the encoder accepts then exits the program
    - -i infile option specifies the input file to encode with Huffman coding. The default input is stdin
    - -o outfile option specifies the output file to write the compressed input to. The default output is stdout
    - -v option specifies to print the compression statistics to stderr. The statistics are the uncompressed file size, the compressed file size and the space saving (100 * (1-(compressed size/uncompressed size))
- The encoder function process::
    - Counts the number of occurrences of each unique symbol in the file and computers a histogram
    - Constructs a Huffman tree using the histogram with a priority queue
    - Creates a code table. The indices in the table represent symbols and the value at each index represents the symbol's code. This step uses a stack of bits to perform a traversal of the Huffman tree.
    - Puts an encoding of the Huffman tree called a tree dump in the file using a post-order traversal of the Huffman tree
    - Go through each symbol in the input file and emit each symbol's code to the output file

Decoder

- The Huffman decoder reads a compressed input file and decompresses it back to its original, uncompressed size. It supports the command line options [-h] [-i infile] [-o outfile] [-v].
    - -h option tells the program to print out a help message describing the purpose of the decoder and the available command-line options then exits the program
    - -i infile specifies the input file to decode using Huffman coding. The default input is stdin
    - -o outfile specifies the output file to write the decompressed input to. The default output is stdout
    - -v prints decompression statistics to stderr. The statistics are compressed file size, the decompressed file size, and space saving (100 * (1 - (compressed size/decompressed size))
- Process for decode/decompress:
    - Reads the tree dump from the input file, using a stack of nodes to reconstruct the Huffman tree
    - Reads in the rest of the file bit-by-bit, traversing down the Huffman tree. A read of 0 leads to walking down the left link while a read of 1 leads to walking down the right link. When a leaf node is reached, its symbol is emitted and the program traverses again from the root.

Pseudocode

node.c

Struct code provided by Professor Long in assignment pdf

```
Typedef struct Node Node;

Struct Node {
        Node *left
        Node *right
        Int symbol
        Int frequency
}

Node *node_create(int symbol, int frequency):
        Node *n=(Node *)malloc(sizeof(Node)
        If (n):
                n->symbol=symbol
                n->frequency=frequency
                n->left=null
                n->right =null
                Return n
        Else:
                free(n)
                Return null
```

```
Void node_delete(Node **n):
        free(*n)
        *n=null

Node *node_join(Node *left, Node *right):
        Return node_create('$', n->left->frequency+n->right->frequency)

Void node_print(Node *n):
        printf(left:<n->left>)
        Print new line
        printf(right:<n->right>)
        Print new line
        printf(symbol:<n->symbol>)
        Print new line
        Printf(frequency:<n->frequency>)
        Print new line
```

Pq.c

```
PriorityQueue *pq_create(int capacity)
        PriorityQueue *q=(PriorityQueue *)malloc(sizeof(PriorityQueue))
        If (q):
                q->capacity=capacity
                q->head=0
                q->tail=0
                q->size=0
                q->elements=(Node *)calloc(capacity,sizeof(Node))
                if(!q->elements):
                        free(q)
                        q=null
        Return q

Void pq_delete(PriorityQueue **q):
        free((*q)->items)
        free(*q)
        *q=null

Bool pq_empty(PriorityQueue *q):
        Return !q->size

Bool pq_full(PriorityQueue *q):
        q->size==q->capacity

Int pq_size(PriorityQueue *q):
        Return q->size
```

```
Bool enqueue(PriorityQueue *q, Node *n):
	If pq_full(q):
		Return false
	q->elements[q->tail]=n
	q->tail=(q->tail+1)%q->capacity
	q->size++
	Return true

Bool dequeue(PriorityQueue *q, Node **n):
	If pq_empty(q):
		Return false
	*n=q->elements[q->head]
	q->head=(q->head+1)%q->capacity
	q->size--
	Return true

Void pq_print(PriorityQueue *q):
	For i in range q->head to q->tail:
		print(q->items[i])
	Print new line
```

Code.c

Struct code provided by Professor Long in assignment pdf

```
Typedef struct Code {
	Int top
	Int bits[MAX_CODE_SIZE]
} Code

Code code_init(void):
	Code *c
	c->top=0
	For i in range MAX_CODE_SIZE:
		c->bits[i]=0
	Return c

Int code_size(Code *c):
	Return c->top

Bool code_empty(Code *c):
	Return !c->top

Bool code_full(Code *c):
	Return c->top==MAX_CODE_SIZE
```

```
Bool code_push_bit(Code *c, int bit):
        If code_full(c):
                Return false
        c->bits[c->top]=bit
        c->top++
        Return true

Bool code_pop_bit(Code *c,int *bit):
        If code_empty(c):
                Return false
        c->top--
        *bit=c->bit[c->top]

Void code_print(Code *c):
        For i in range c->top:
                Print <c->bits[i]>
```

Io.c

```
Int read_bytes(int infile, int *buf, int nbytes):
        Int n=0
        While (n<nbytes and read(infile,buf,1)>0):
                n++
        Return n

Int write_bytes(int outfile, int *buf, int nbytes):
        Int n=0
        while(n<nbytes and write(infile,buf,1)>0):
                n++
        Return n
```

Influenced by Eugene’s lab section on 5/11

```
Bool read_bit(int infile, int *bit):
        If buffer is empty:
                Fill it
        *bit=buffer[bufindex]
        bufindex++
        If (bufindex==BLOCK*8):
                Return false
        Return true
```

Inspired by Eugene’s lab section on 5/11

```
Void write_code(int outfile, Code *c)
        For i in range c->top:
                If get_bit(c,i)==1:
```

```
                set_bit(buffer, bufindex)
            Else:
                clr_bit((buffer, bufindex)
        Bufindex++
        If bufindex==BLOCK*8:
            write_bytes(outfile,buffer,BLOCK)
            bufindex=0

Void flush_codes(int outfile):
        If bufindex>0:
            Int offset=bufindex % BLOCK ? 1 : 0
            write_bytes(outfile,buffer, bufindex+offset)
```

stack.c

Struct code provided by Professor Long in assignment pdf

```
Struct Stack {
        Int top
        Int capacity
        Node **items
}

Stack *stack_create(int capacity):
        Stack *s=(Stack *s) malloc(sizeof(Stack))
        If (s):
            s->top=0
            s->capacity=capacity
            s->items=(Node **)calloc(capacity,sizeof(Node *))
            If (!s->items):
                free(s)
                s=null
        Return s

Void stack_delete(Stack **s):
        if(*s and (*s)->items):
            free((*s)->items)
            free(*s)
            *s=null

Bool Stack_empty(Stack *s):
        Return !s->top

Bool stack_full(Stack *s):
        Return s->top==s->capacity
```

```
Int stack_size(Stack *s):
        Return s->top

Bool stack_push(Stack *s, Node *n):
        if(stack_full(s)):
                Return false
        s->items[s->top]=*n
        s->top++
        Return true

stack_pop(Stack *s, Node **n):
        if(stack_empty(s)):
                Return false
        s->top--
        **n=s->items[s->top]
        Return true

stack_print(Stack *s):
        For i in range s->top:
                Print <s->items[i]>
        Print new line
```

Huffman.c

Code inspired by Eugene's 5/11 lab section

```
postorder(Node *n):
        If n!=null:
                postorder(n->left)
                postorder(n->right)
                node_print(n)
        Store tree dump

Node *build_tree(int hist[static ALPHABET]):
        Create nodes from histogram
        Create priority queue
        Put nodes into priority queue
        While pq_size>1:
                dequeue to get left child
                Dequeue to get right child
                Parent = node_join(left child, right child)
                Enqueue parent
        Return dequeued root

Void build_codes(Node *root, Code table[static ALPHABET]):
        For each symbol in Huffman tree:
```

```
            Create a code
            If left child:
                Push 0 onto the code
            If right child:
                Push 1 onto the code
            table[symbol]=code

Node *rebuild_tree(int bytes, int tree_dump[static nbytes]):
    postorder()
    Create histogram
    Return build_tree(histogram)

Void delete_tree(Node **root):
    postorder()
    free(tree dump)
    **root=null
```

Draft Work

priority queue (5 elements)

slot

[2 , 3 , 4 , 5 , 6]

head tail

5, 3, 2, 4, 6

pq[slot-1] > 6

pq[slot] = pq[slot-1]

right(k, n)

(k+1)% n

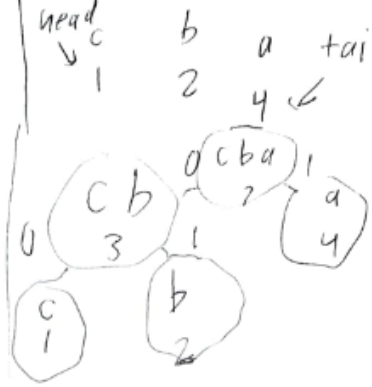
input aaaabbc

symbol	freq
a	4
b	2
c	1

head

head c b a tail

1 2 4



symbol	code
a	1
b	01
c	00

a	4
b	2
c	1
d	1

symbol	code
a	0
b	11
c	100
d	101

input: a a a a b b c d

a a a a b b c d

0 0 0 0 11 11 100 101

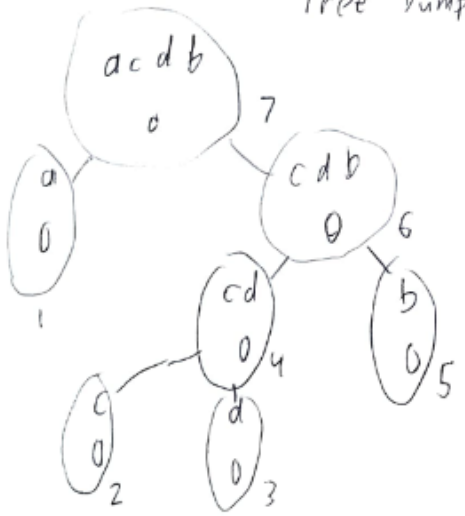
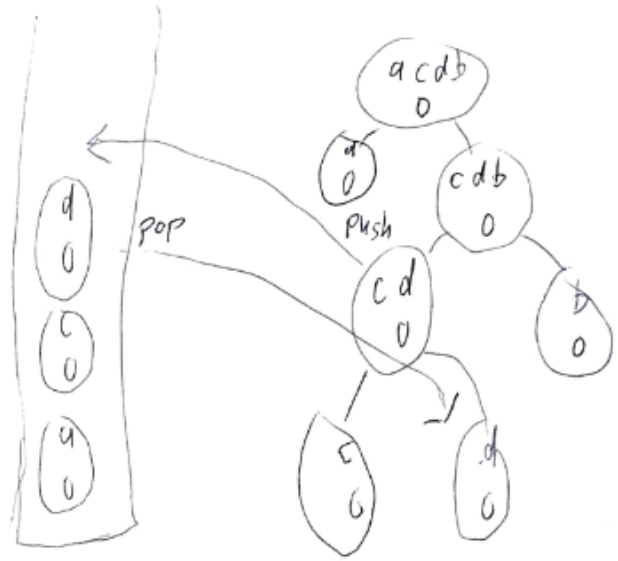
if both children are NULL, then it's a leaf

postorder traversal

```
def postorder(Node *n):
    if n is not null{
        post order(n->left)
        postorder(n->right)
        node_print(n)
    }
```

9, 10, 5, 7, 8, 6, 3

Tree Dump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